

A Reflection of My General Education Courses at Iowa State University

When I was a kid, I had heard of the term “Engineer” before, but I never quite knew what it meant. When I finally had the courage to ask someone, I was told that engineers solve problems to make the world a better place. Although this may be true, it leaves out important details. There is a lot of work that goes into figuring out which problems to solve and how to formulate a solution that caters to others. After all, a solution is only feasible if people can understand it, afford it, and be willing to use it. The technical classes I have taken at Iowa State have taught me how to build things, but my general education classes taught me to think about the people those things are built for. They didn’t improve my coding or software management skills. Instead, my general education classes changed the questions I ask before I begin designing solutions.

One of the most important changes was learning to question my own assumptions. In INTST 2500, we learned about different regions of the world, focusing on the climates, economies, and cultures that made each region unique. I found that many of the regions were completely different from where I grew up. I felt very similarly when I took RUS 3750, which explored Russia from Gorbachev to Putin. Specifically, I was shocked how quickly the Soviet Union fell apart and was fascinated that most people in Russia today cannot protest the government out of fear of being put in jail. This definitely gave me an appreciation for the freedom we have in the USA. I take speaking up and expressing my feelings for granted, and RUS 3750 showed me that what feels normal for me is not normal everywhere. HDFS 2760: Human Sexuality drove this point home. In HDFS 2760, we studied how gender expectations and culture as a whole shape the way people think and make choices. I was really surprised how often people view their own norms as universal. In actuality, the world is very diverse. As an engineer, this is a very important insight. If a design team assumes every user thinks exactly like them, they’ll end up building something that works well for some people but ignores the needs of another user group. Through these classes, I have learned to try to gather more than one perspective before designing a solution, even if it takes more time.

Another lesson that stuck with me through my general education classes is that the most obvious solution isn’t always the right one. In Russia 3750 the one topic that I vividly remember was Chernobyl. After the nuclear accident, people were forced to leave their homes because of the dangerous radiation levels. Despite the health risk and directives from the government, some of the older generation still live inside the contamination zone. They refused to leave because they were at peace in the place they’ve always called home and weren’t willing to move, even if it meant more safety. Again, INTST 2500 echoed a similar message about the complexity of seemingly simple solutions. Specifically, I noticed

the complexity of the ivory smuggling problem in Africa. Initially, I assumed that it was a simple poaching problem. What I did not realize was the governments there run entire task forces to stop it. I was shocked to learn that officials even found a whole shipping container full of tusks. This problem involves local law enforcement, international shipping, and a lot of money. Both of these classes taught me that real problems are often not as simple as they first appear, and that a solution has to account for the people, politics and economics around it.

Another skill I gained is working with constraints. In HDFS 2830: Personal Finance, we learned calculations like debt-to-income ratios, future value, and interest rates. Personally, I learned the most from the case studies, where we had to apply the concepts learned in class to real-world scenarios. Working through these case studies helped me feel more prepared for my own life and even changed how I think about cost. Cost is much more complicated than it seems. Something that appears cheap up front can end up costing more over time when factoring in repairs and replacements. And price is important. If the people who need a solution can't afford it, it doesn't matter how well it was engineered. Almost every engineering project has some sort of constraint. Whether it is a budget, a deadline, or a regulation to work around, there are always limitations to project. Because of my general education classes, I'm more comfortable now making decisions inside those limits instead of getting stuck on what I can't change.

These lessons matter even more when I look at the problems engineers face today. For example, look at the global supply chains. A shutdown or shipping delay in one part of the world can raise prices in another part of the world. After INTST 2500 and RUS 3750, I understand why. It is because regions rely on each other for resources. Decisions made by global leaders can change things overnight. I no longer think relying on one supplier or country is a viable option. Having a backup plan is important. Another problem that highlights this is the rising costs around the world. HDFS 2830 taught me to look at what a solution costs over its whole lifetime, not just when it is purchased. Additionally, it has helped me think about what people with limited incomes can realistically afford. In software, one issues I am most likely to face is the digital divide. If I build a website that needs fast internet or a newer device, rural and low income users cannot benefit from my solution. HDFS 2760 showed me that the best way to avoid these mistakes is to ask people what they need instead of assuming I already know.

Overall, my general education classes didn't make me a better programmer, but they did make me a more careful engineer. Before taking these classes, I thought of a problem as something to solve as quickly and cleanly as possible. Now I see that every problem has people involved in some way. These people have different backgrounds,

budgets, and beliefs. Learning about Russia, world regions, human sexuality, and personal finance gave me a habit of slowing down to ask who I am building for and what they actually need. I know my code will still be judged on whether it works. But I want it to work for the people who use it, not just the ones who think like me. That is how I would answer the question I asked as a kid. Engineers solve problems, and the good ones start by understanding who has them.